

Generating Functions

Section 2.6.1–2.6.3

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



$$2 + 3 + 2 = 7$$

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Today's lecture

1. Double Decks

2. Counting with Repetition

3. Changing Money

4. Examples and Exercises

Section 2.6.1: Double decks

Question

Suppose two identical copies of each card type are available. How many different hands of a prescribed size can be formed?

We record card types and multiplicities, not which physical copy was selected. Ordinary binomial coefficients alone do not account for these repeated types.

A small double deck

Example

With two copies each of A, B, and C, the two-card hands are

AA, AB, AC, BB, BC, CC.

Thus there are 6 hands, not $\binom{6}{2} = 15$. The latter treats the two copies of each type as distinguishable.

A polynomial for one card type

For each type, choose zero, one, or two copies. Record these choices by

$$1 + x + x^2.$$

The exponent records how many cards are selected. The coefficient records how many ways that multiplicity can occur in our model.

Multiplying the choices

For three card types,

$$(1 + x + x^2)^3 = 1 + 3x + 6x^2 + 7x^3 + 6x^4 + 3x^5 + x^6.$$

The coefficient of x^k counts hands of size k .

For example, the coefficient 6 of x^2 counts the six hands listed earlier.

Ordinary generating functions

Definition

The ordinary generating function of a sequence $(a_n)_{n \geq 0}$ is the formal power series

$$A(x) = \sum_{n \geq 0} a_n x^n.$$

We write $[x^n]A(x) = a_n$ for coefficient extraction.

Formal power series

Here x is a bookkeeping symbol. We compare coefficients rather than evaluate the series at a real number.

An infinite series is useful when the coefficient of each fixed power can be determined by finitely many relevant choices.

Analytic convergence is a separate question.

Coefficient rules

If $A(x) = \sum a_n x^n$ and $B(x) = \sum b_n x^n$, then

$$[x^n](A + B) = a_n + b_n,$$

$$[x^n](AB) = \sum_{j=0}^n a_j b_{n-j}.$$

The second formula is the convolution rule.

Why products count combined choices

Proof

To form an object of total size n from an A-object and a B-object, choose their sizes j and $n - j$. For that split there are $a_j b_{n-j}$ choices. Sum over all splits.

Multiplying generating functions combines independent choices and adds their sizes.

The general double-deck answer

Formula

For m card types with at most two copies of each, the number of hands of size k is

$$[x^k](1 + x + x^2)^m.$$

Each factor corresponds to one distinguishable type.

Counting by the number of pairs

Equivalent formula

The same number is

$$\sum_{j=0}^{\lfloor k/2 \rfloor} \binom{m}{j} \binom{m-j}{k-2j},$$

with impossible choices interpreted as zero.

Proof

Choose the j types appearing twice, then the $k - 2j$ types appearing once. Every hand has a unique number of pairs.

Five cards from two identical standard decks

There are 52 card types. A five-card hand contains zero, one, or two pairs of identical types. Hence its count is

$$\binom{52}{5} + 52 \binom{51}{3} + \binom{52}{2} \binom{50}{1}.$$

This equals $[x^5](1 + x + x^2)^{52}$.

Symmetry of the coefficients

Observation

$$[x^k](1 + x + x^2)^m = [x^{2m-k}](1 + x + x^2)^m.$$

Proof

Replace each multiplicity $a_i \in \{0, 1, 2\}$ by $2 - a_i$. A hand of size k becomes a hand of size $2m - k$, and the operation reverses itself.

Section 2.6.2: Counting with repetition

Question

How many multisets of size n can be chosen from r distinguishable types, with unlimited repetition?

Record the multiplicities $a_1, \dots, a_r$. The problem is to count

$$a_1 + \cdots + a_r = n, \quad a_i \geq 0.$$

Stars and bars

$$\begin{array}{ccccc|ccc|cc} \star & \star & & & & \star & \star & \star & & & \star & \star \\ 2 & + & 3 & + & 2 & = & 7 \end{array}$$

Represent n objects by stars and separate the r types by $r - 1$ bars. Empty groups are allowed.

The stars-and-bars formula

Theorem

The number of nonnegative solutions of $a_1 + \cdots + a_r = n$ is

$$\binom{n+r-1}{r-1} = \binom{n+r-1}{n}.$$

Proof

A solution is uniquely encoded by n stars and $r-1$ bars. Choose the positions of the bars in a string of length $n+r-1$.

The geometric series

$$1 + x + x^2 + \cdots = \frac{1}{1 - x}.$$

Proof

Multiplication by $1 - x$ cancels every positive-degree term and leaves 1. This is a formal identity.

Unlimited copies of one type contribute the factor $(1 - x)^{-1}$.

The generating function for repetition

Identity

$$\frac{1}{(1-x)^r} = \sum_{n \geq 0} \binom{n+r-1}{r-1} x^n.$$

Proof

The coefficient in the product of r geometric series counts nonnegative solutions with total n . Apply stars and bars.

Positive multiplicities

Formula

For $n \geq r$, the number of positive solutions of $a_1 + \cdots + a_r = n$ is

$$\binom{n-1}{r-1}.$$

Proof

Subtract 1 from every coordinate, leaving a nonnegative total of $n - r$. Equivalently, use generating function

$$\left(\frac{x}{1-x} \right)^r.$$

Different lower bounds

Question

Count solutions of $a + b + c = 12$ with $a \geq 2$, $b \geq 3$, $c \geq 0$.

Set $a' = a - 2$ and $b' = b - 3$. Then $a' + b' + c = 7$, so the answer is

$$\binom{9}{2} = 36.$$

The generating function is $x^5/(1-x)^3$.

Generating function

The number of solutions of $a_1 + \cdots + a_r = n$ with $0 \leq a_i \leq b_i$ is

$$[x^n] \prod_{i=1}^r (1 + x + \cdots + x^{b_i}).$$

Each factor lists exactly the allowed values of one coordinate.

Recovering inclusion and exclusion

When every upper bound is b ,

$$(1 + x + \cdots + x^b)^r = \frac{(1 - x^{b+1})^r}{(1 - x)^r}.$$

Expand the numerator by the binomial theorem and the denominator by stars and bars. The coefficient of x^n is

$$\sum_{j=0}^{\min(r, \lfloor n/(b+1) \rfloor)} (-1)^j \binom{r}{j} \binom{n - j(b+1) + r - 1}{r - 1}.$$

Parity restrictions

Example

Count solutions of $a + b + c = 8$ with a even and $b, c \geq 0$.

Use

$$\frac{1}{(1 - x^2)(1 - x)^2}.$$

Summing over $a = 0, 2, 4, 6, 8$ gives $9 + 7 + 5 + 3 + 1 = 25$.

Section 2.6.3: Changing money

Question

How many ways can a sum be formed using coins of specified denominations?

Coins of the same denomination are indistinguishable. Order does not matter.

A denomination d contributes

$$1 + x^d + x^{2d} + \dots = \frac{1}{1 - x^d}.$$

The coin generating function

Formula

For unlimited coins of denominations $d_1, \dots, d_r$, the number of ways to form amount n is

$$[x^n] \prod_{i=1}^r \frac{1}{1 - x^{d_i}}.$$

The exponent records value, rather than the number of coins.

Coins of values 1, 2, and 5

Question

How many ways can these coins make 10 units?

Solve $a + 2b + 5c = 10$ in nonnegative integers.

c	Remaining value	Choices for b
0	10	6
1	5	3
2	0	1

Total: $6 + 3 + 1 = 10$.

A coefficient version of the calculation

$$[x^{10}] \frac{1}{(1-x)(1-x^2)(1-x^5)} = 10.$$

With only values 1 and 2, amount m can be formed in $\lfloor m/2 \rfloor + 1$ ways.

Sum this count for remaining amounts 10, 5, 0 after choosing the number of 5-coins.

Several denominations

For denominations 1, 5, 10, 25, 50, 100, the generating function is

$$\frac{1}{(1-x)(1-x^5)(1-x^{10})(1-x^{25})(1-x^{50})(1-x^{100})}.$$

Only terms up to the desired amount are needed when computing a coefficient.

A recurrence for changing money

Notation

Let $f_j(n)$ count ways to form n using the first j denominations. Set $f_j(n) = 0$ for $n < 0$.

Recurrence

$$f_j(n) = f_{j-1}(n) + f_j(n - d_j).$$

The two terms count choices using no d_j -coin and choices using at least one d_j -coin.

Initial conditions

$$f_0(0) = 1, \quad f_0(n) = 0 \quad (n > 0).$$

The empty collection is the unique way to form zero. These initial conditions and the recurrence determine all counts.

Removing one d_j -coin from a collection in the second case explains why that term still allows all j denominations.

Does order matter?

Question

How many ways can 4 be written using parts 1 and 2?

Unordered: $1 + 1 + 1 + 1$, $2 + 1 + 1$, $2 + 2$, giving 3.

Ordered: 1111, 112, 121, 211, 22, giving 5.

The product $1/((1-x)(1-x^2))$ counts the unordered version.

Question

Find the number of solutions of $a + b + c = 10$ with $a \geq 1$, $0 \leq b \leq 3$, and c even.

Write a generating function, then count directly by the value of c .

Practice: solution

The generating function is

$$\frac{x(1 + x + x^2 + x^3)}{(1 - x)(1 - x^2)}.$$

For $c = 0, 2, 4, 6$, each of $b = 0, 1, 2, 3$ gives a positive a . For $c = 8$, only $b = 0, 1$ work. For $c = 10$, none work.

$$4 + 4 + 4 + 4 + 2 = 18.$$

- A generating function records counts as coefficients.
- Products combine independent choices and add sizes.
- Repetition and bounds are encoded by geometric-series factors.
- Coin problems use exponents for total value and distinguish order from multiplicity.

Fibonacci Numbers and Recurrences